

AWS CLOUD PRACTITIONER PROJECT

Hands-on AWS Services Demonstration & Evidence Documentation

Prepared for GitHub Portfolio Upload

1. Project Overview

This project documents hands-on exploration of AWS cloud services through the AWS Management Console and related developer tools. The screenshots provide visual evidence of work involving web hosting, storage lifecycle management, analytics, compute, content delivery, generative AI, NLP, computer vision and voice application testing.

2. Objectives

- Understand and practice core AWS cloud services.
- Explore compute, storage, analytics and content-delivery workflows.
- Experiment with AWS AI/ML services such as Bedrock, Comprehend and Rekognition.
- Understand how cloud services can support real-world application development.
- Maintain organized screenshot evidence suitable for a GitHub portfolio.

3. Services and Activities

#	Service / Activity	Demonstration
1	HTML/CSS Web Interface	Navigation bar and dashboard-style menu.
2	Amazon S3	Lifecycle transitions and object expiration.
3	Amazon Athena	SQL query results for e-commerce data.
4	Amazon CloudFront	Website accessed through a CloudFront distribution.
5	Amazon EC2	Instance states, types, zones and status checks.
6	Alexa Developer Console	Alexa skill testing using the simulator.
7	Amazon Bedrock	Generative AI prompt and response testing.
8	Amazon Rekognition	Text detection from an image.
9	Amazon Comprehend	Key-phrase extraction from text.
10	Amazon Rekognition	Image properties and dominant-color analysis.

4. Project Flow

The practical activities can be viewed as a collection of cloud building blocks: compute and hosting with EC2, object storage with S3, data querying with Athena, content delivery with CloudFront, and AI/ML experimentation with Bedrock, Alexa, Comprehend and Rekognition.

5. Detailed Evidence

Web Navigation Bar

A simple HTML/CSS interface containing Home, About, Services and Contact navigation items, along with a dashboard-style menu containing Dashboard, Profile, Settings and Logout.

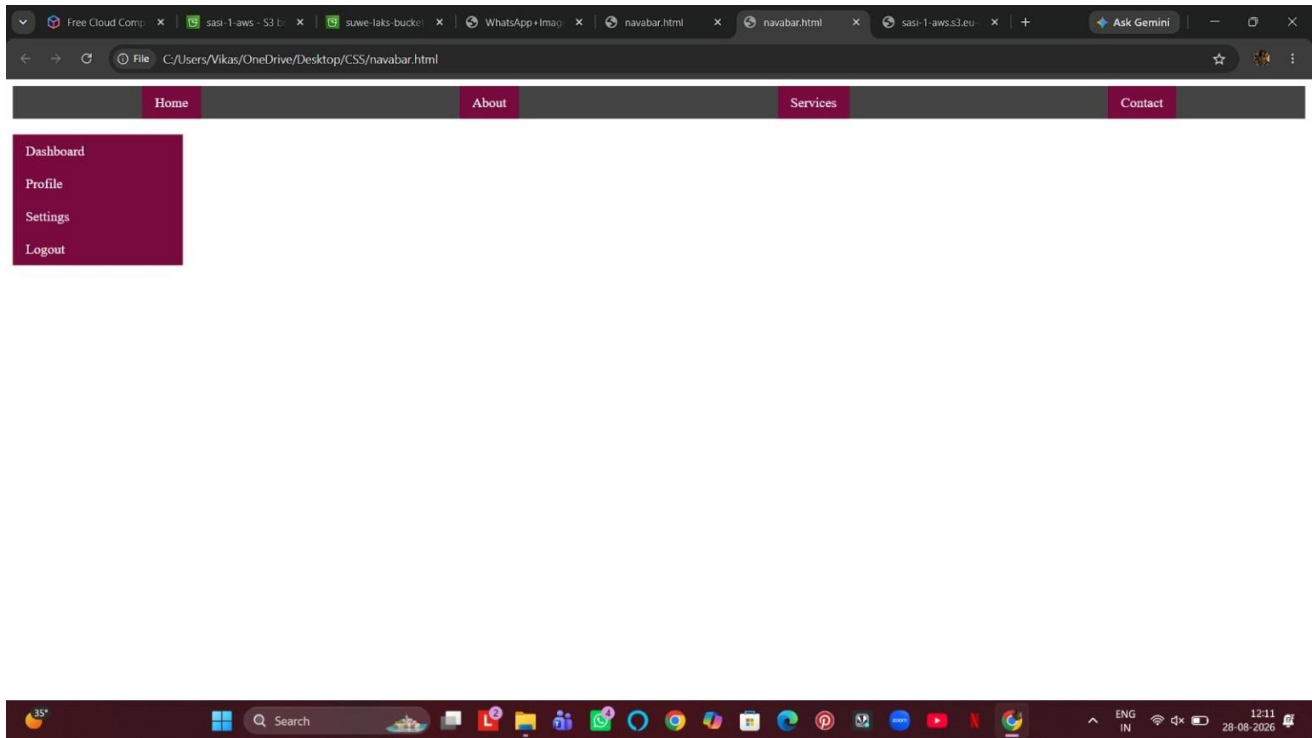


Figure 1: Web Navigation Bar

Amazon S3 Lifecycle Configuration

The lifecycle rule demonstrates automated storage management. Objects are uploaded on Day 0, transition to Standard-IA on Day 10, transition to One Zone-IA on Day 30, and are configured to expire later. This helps reduce storage cost and automate retention.

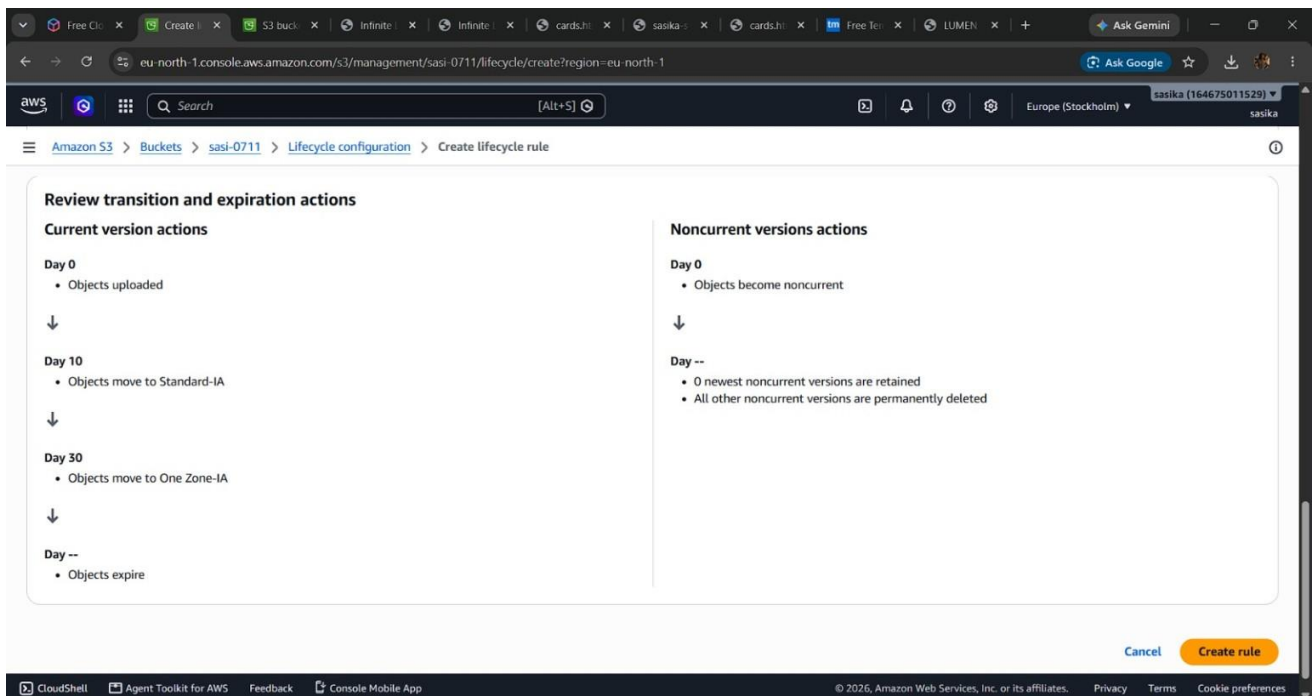
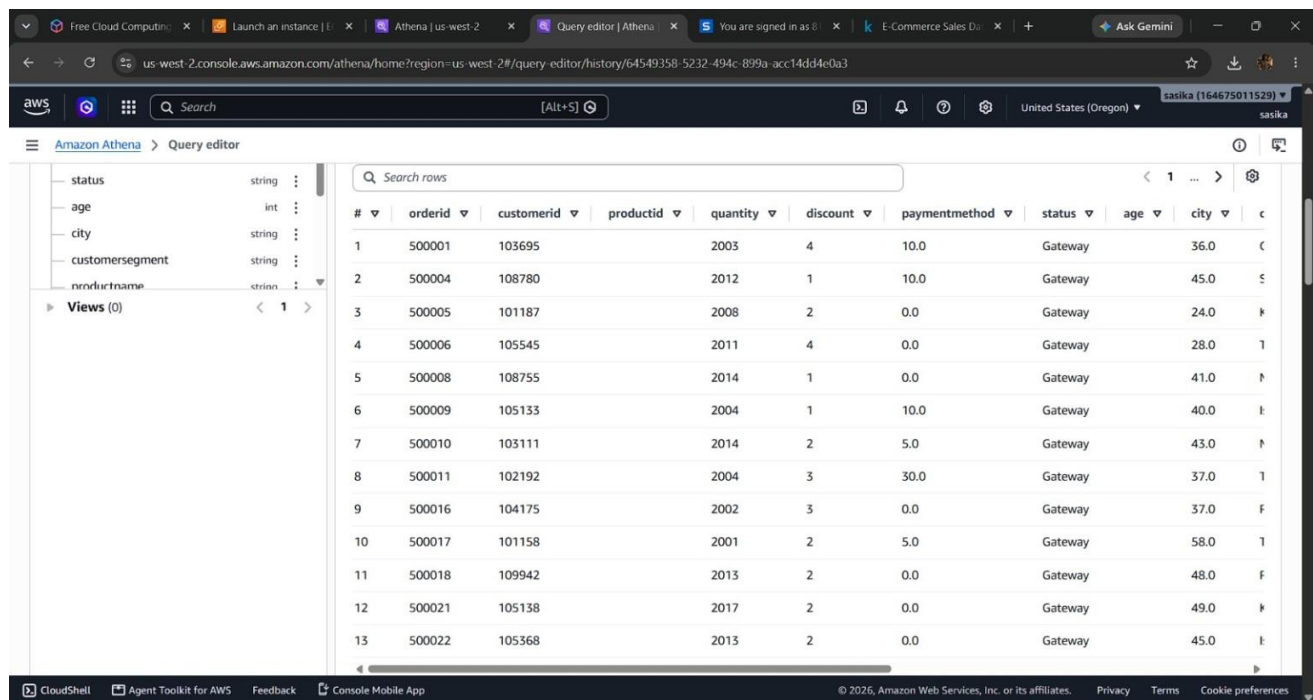


Figure 2: Amazon S3 Lifecycle Configuration

Amazon Athena Query Results

The Athena query editor displays an e-commerce/order dataset. Visible columns include order ID, customer ID, product ID, quantity, discount, payment method, status, age and city.



The screenshot shows the Amazon Athena Query editor interface. On the left, there's a sidebar with a schema tree showing columns like status, age, city, customersegment, and productname. The main area displays a table with 13 rows of data. The columns are: #,orderid, customerid, productid, quantity, discount, paymentmethod, status, age, city, and c. The data includes various order IDs, customer IDs, product IDs, quantities, discounts, payment methods (all Gateway), statuses (all Gateway), ages, and cities.

#	orderid	customerid	productid	quantity	discount	paymentmethod	status	age	city	c
1	500001	103695	2003	4	10.0	Gateway	Gateway	36.0	C	
2	500004	108780	2012	1	10.0	Gateway	Gateway	45.0	S	
3	500005	101187	2008	2	0.0	Gateway	Gateway	24.0	F	
4	500006	105545	2011	4	0.0	Gateway	Gateway	28.0	T	
5	500008	108755	2014	1	0.0	Gateway	Gateway	41.0	F	
6	500009	105133	2004	1	10.0	Gateway	Gateway	40.0	I	
7	500010	103111	2014	2	5.0	Gateway	Gateway	43.0	F	
8	500011	102192	2004	3	30.0	Gateway	Gateway	37.0	T	
9	500016	104175	2002	3	0.0	Gateway	Gateway	37.0	F	
10	500017	101158	2001	2	5.0	Gateway	Gateway	58.0	T	
11	500018	109942	2013	2	0.0	Gateway	Gateway	48.0	F	
12	500021	105138	2017	2	0.0	Gateway	Gateway	49.0	F	
13	500022	105368	2013	2	0.0	Gateway	Gateway	45.0	I	

Figure 3: Amazon Athena Query Results

Amazon CloudFront Hosted Website

The browser shows a deployed website being served through a CloudFront distribution. The page presents the LUMEN // 80 furniture and spatial-design interface.

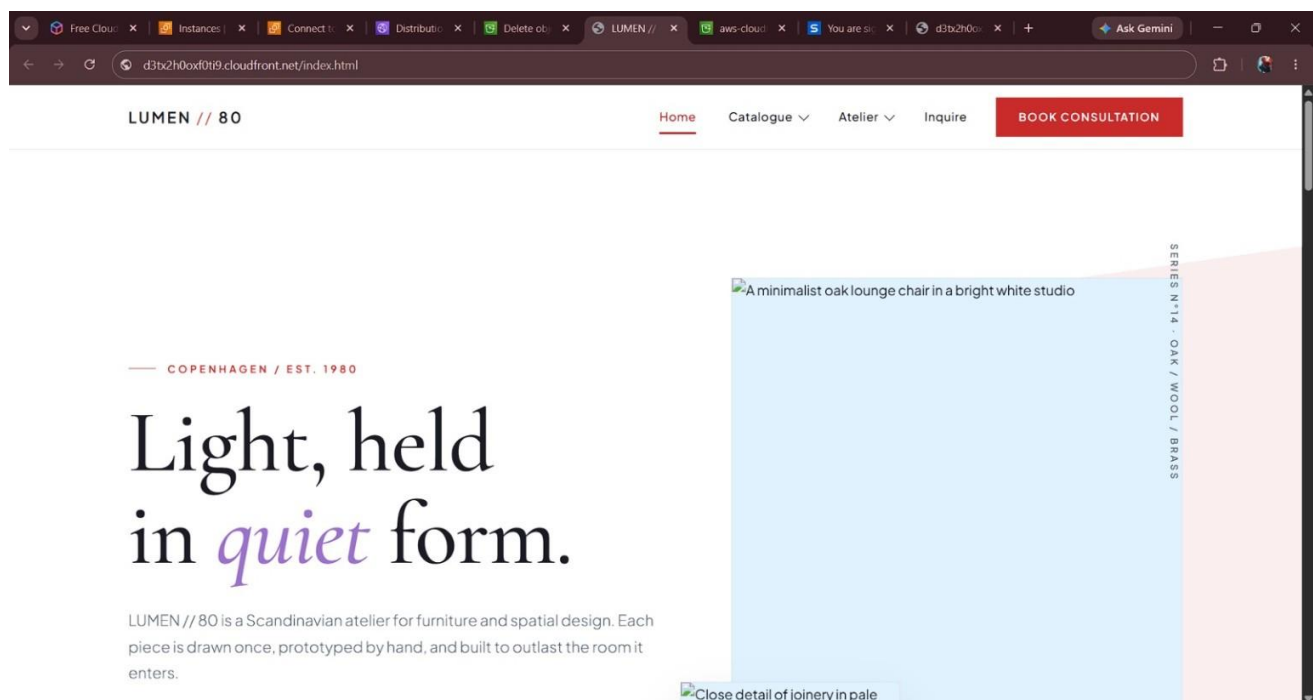


Figure 4: Amazon CloudFront Hosted Website

Amazon EC2 Instances

The EC2 console shows several instances in Running or Terminated states. The view also displays instance IDs, instance types, status checks and availability zones.

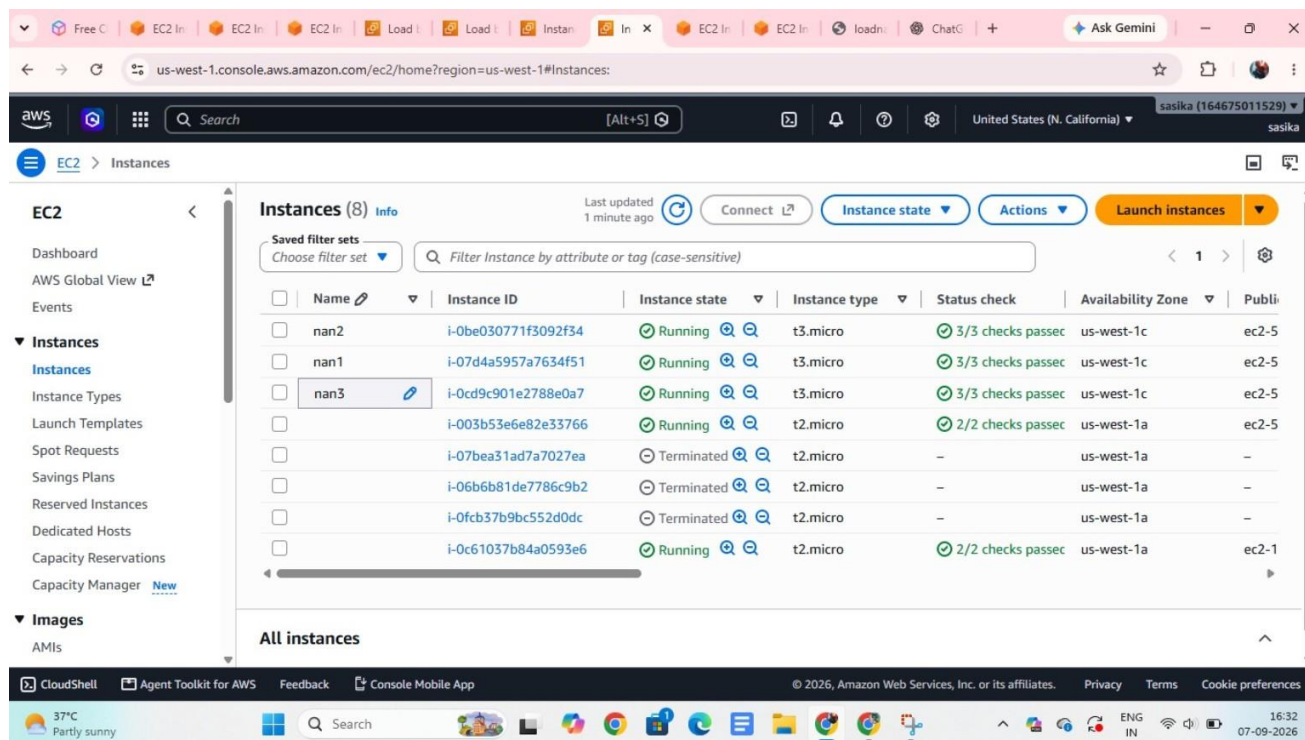


Figure 5: Amazon EC2 Instances

Alexa Developer Console

The Alexa Developer Console simulator is used to test a skill. The screenshot shows the testing environment, skill I/O, device display and an example response.

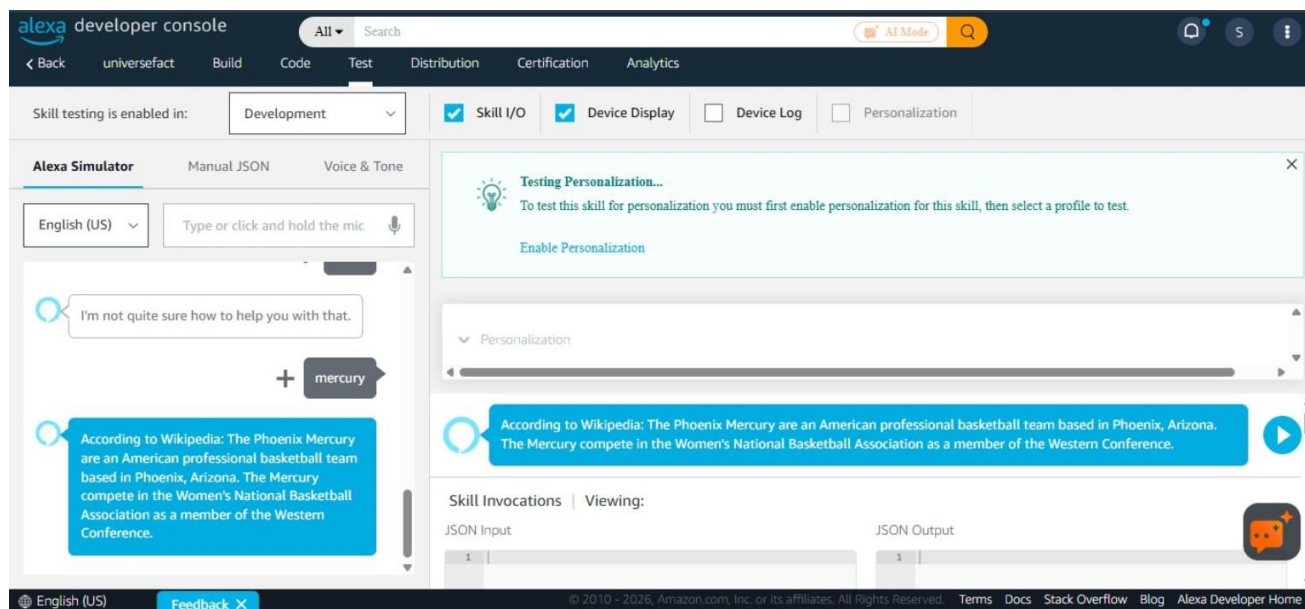


Figure 6: Alexa Developer Console

Amazon Bedrock Playground

The Bedrock Playground is used to test a foundation model with a prompt and generated response. The screenshot demonstrates managed generative-AI experimentation inside AWS.

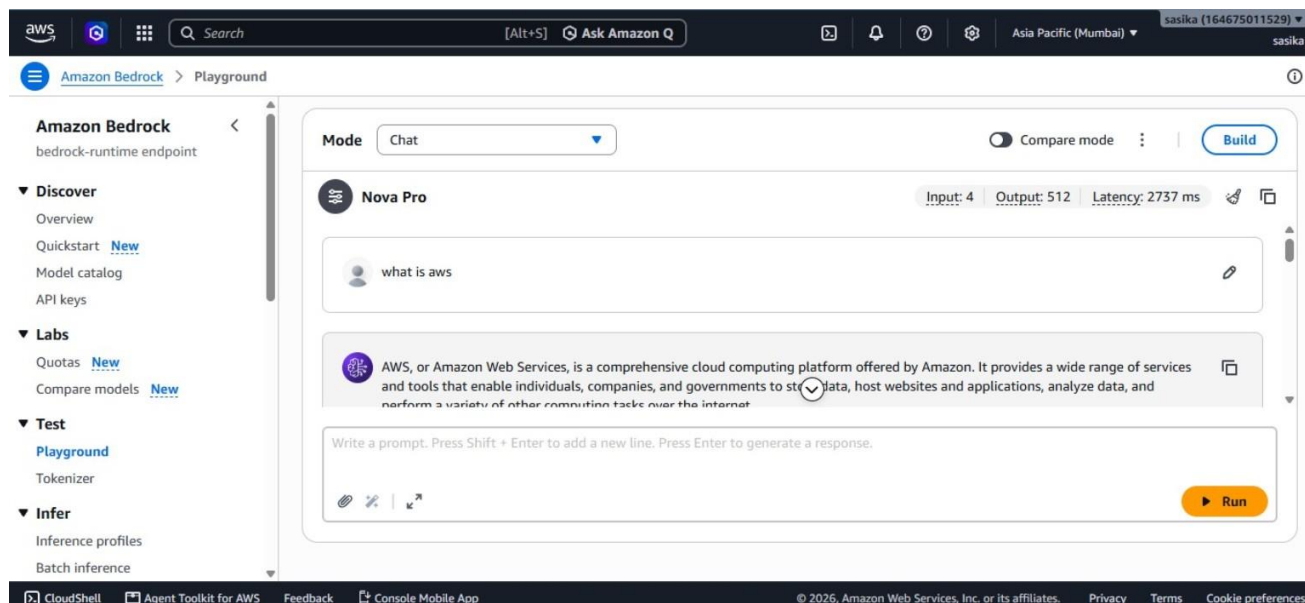


Figure 7: Amazon Bedrock Playground

Amazon Rekognition – Text Detection

Rekognition detects text appearing in an image. The example returns detected words such as FIND, BALANCE and IN LIFE with result information.

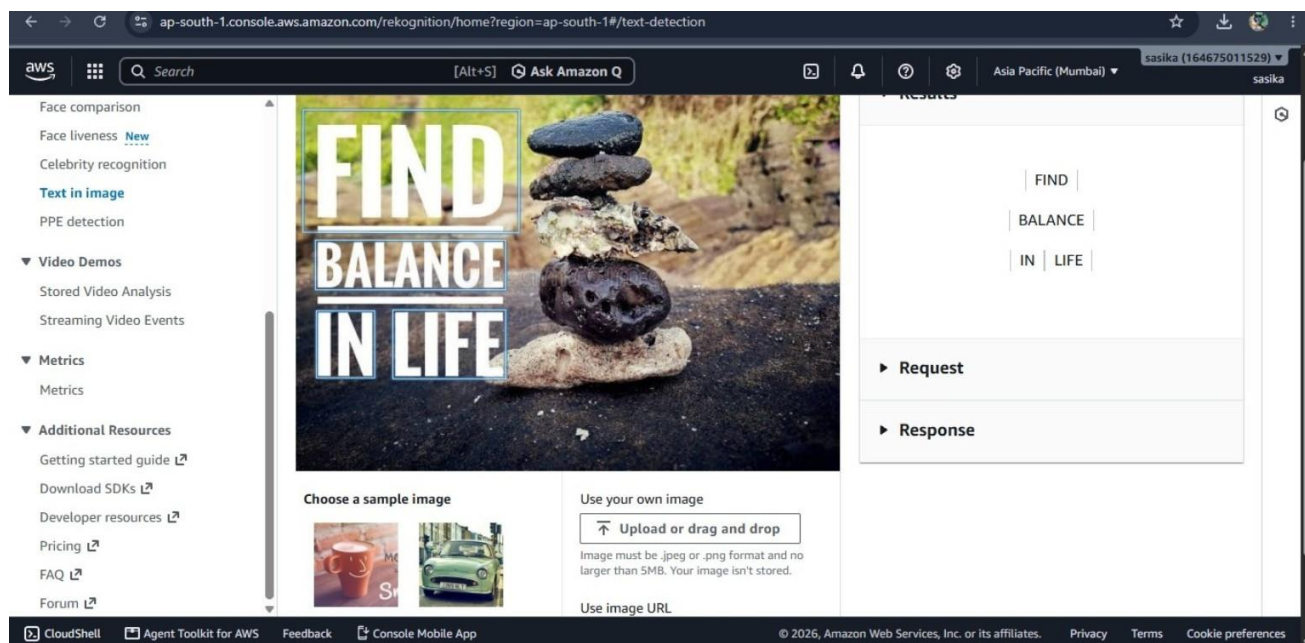


Figure 8: Amazon Rekognition – Text Detection

Amazon Comprehend – Key Phrase Analysis

Comprehend analyzes natural-language text and extracts key phrases with confidence scores. The example analyzes an office-network troubleshooting paragraph.

The screenshot displays the Amazon Comprehend console interface. The left sidebar contains navigation options: Real-time analysis, Analysis jobs, Customization (Custom classification, Custom entity recognition, Flywheels, Endpoints), and a search bar. The main content area shows a text input field with a paragraph about office network issues. Below the input, the 'Results' section displays a table of key phrases and their confidence scores.

Key phrases	Confidence
The office network	0.99+
a router	0.99+

Figure 9: Amazon Comprehend – Key Phrase Analysis

Amazon Rekognition – Image Properties

Rekognition analyzes visual properties of an image, including dominant colors and image-quality information such as brightness.

The screenshot displays the Amazon Rekognition console interface. The left sidebar contains navigation options: Demos (Label detection, Image properties, Image moderation, Facial analysis, Face comparison, Face liveness, Celebrity recognition, Text in image, PPE detection), Video Demos (Stored Video Analysis, Streaming Video Events), Metrics, and Additional Resources. The main content area shows the 'Image Properties' section with a large image of a woman at a market. To the right, the 'Results' section displays a table of dominant colors and image quality metrics.

Dominant Colors		
#2f4f4f	RGB(47, 79, 79)	30.89%
#d2b48c	RGB(210, 180, 140)	16.79%
#cd853f	RGB(205, 133, 63)	11.60%
#808080	RGB(128, 128, 128)	11.05%
#708090	RGB(112, 128, 144)	8.913%

Image Quality	
Brightness	74.93

Figure 10: Amazon Rekognition – Image Properties

6. Key Learning Outcomes

The project demonstrates practical familiarity with the AWS console and with several major cloud service categories. It also shows how AWS can be used across the application lifecycle—from hosting and storage to querying, delivery and intelligent image/text processing.

- Cloud compute: EC2
- Object storage and lifecycle management: S3
- Serverless SQL analytics: Athena
- Content delivery: CloudFront
- Generative AI: Amazon Bedrock
- Voice/skill development: Alexa Developer Console
- Natural-language processing: Amazon Comprehend
- Computer vision and OCR: Amazon Rekognition

7. GitHub Repository Structure

Keep the README.md at the repository root and store the evidence images under screenshots/. The full Word documentation is kept under docs/.

```
aws-cloud-practitioner-project/  
├── README.md  
├── docs/  
│   └── AWS_Cloud_Practitioner_Project_Documentation.docx  
└── screenshots/  
    ├── 01_Web_Navigation_Bar.jpg  
    ├── 02_S3_Lifecycle_Configuration.jpg  
    ├── 03_Amazon_Athena_Query_Results.jpg  
    ├── 04_Amazon_CloudFront_Website.jpg  
    ├── 05_Amazon_EC2_Instances.jpg  
    ├── 06_Alexa_Developer_Console.jpg  
    ├── 07_Amazon_Bedrock_Playground.jpg  
    ├── 08_Amazon_Rekognition_Text_Detection.jpg  
    ├── 09_Amazon_Comprehend_Key_Phrases.jpg  
    └── 10_Amazon_Rekognition_Image_Properties.jpg
```

8. Public GitHub Safety Checklist

Before making the repository public, check the screenshots and files for information that should not be exposed publicly.

- AWS account IDs and account-specific identifiers
- IAM usernames or internal organization details
- Access keys, secret keys, tokens or passwords
- Private URLs, endpoints or internal hostnames
- Personally identifiable information

- Any dataset that is not licensed for public sharing